# HRSU Indore Pvt. Ltd.

Beyond Granules.
The Purity of Powder.

Ref: HRSU/TDS/CNT/001 Date: 28 September 2026

## TECHNICAL DATA SHEET

**Calcium Nitrate Tetrahydrate - Industrial Grade (Crystalline Powder)**

### 1. Product Information

| | | | |
|---|---|---|---|
| **Product Name** | Calcium Nitrate Tetrahydrate | **Packaging** | 25 kg / 50 kg PP woven bag with inner PE liner |
| **CAS No.** | 13477-34-4 | **HS / HSN Code** | 28342990 |
| **EINECS EC No.** | 233-332-1 | **Molecular Formula** | $Ca(NO_3)_2 \cdot 4H_2O$ |
| **Molecular Weight** | 236.15 g/mol | **UN No. / Class** | UN 1454 / Class 5.1 (Packing Group III) |
| **Appearance** | White crystalline powder | **Supply / Samples** | Commercial bulk & plant trial samples available |

### 2. Technical Specifications

| Sr. No. | Parameter | Specification |
|---|---|---|
| 1 | Appearance | **White Crystalline Powder** |
| 2 | Solubility in Water | **Hygroscopic, Easily & Completely Soluble in Water** |
| 3 | Assay (as $Ca(NO_3)_2 \cdot 4H_2O$) | **98.0% Min** |
| 4 | Active $Ca(NO_3)_2$ Content (Dry Salt Basis) | **68.0% Min (~69.5% Theoretical)** |
| 5 | Calcium (as Soluble Ca) | **15.5 – 17.0%** |
| 6 | Nitrate Nitrogen (as N) | **11.5 – 12.0%** |
| 7 | pH (5% w/v Aqueous Solution) | **4.5 – 7.0** |
| 8 | Water Insoluble Matter | **0.15% Max** |
| 9 | Free Moisture | **2.0% Max** |

***Composition Note:** Pure Calcium Nitrate Tetrahydrate ($Ca(NO_3)_2 \cdot 4H_2O$) consists of ~69.5% active $Ca(NO_3)_2$ and ~30.5% chemically bound water of crystallisation. Unlike agricultural prills, it is uncoated (zero wax scum) and free of ammoniacal nitrogen.

### 3. Applications & Wastewater Dosing Guidelines

- **Industrial Applications:** Sewage & Wastewater Treatment ($H_2S$ & odour control, anoxic biogas scrubbing, sludge conditioning), Concrete Accelerators, Rubber/Latex Coagulants, and Mining/Oilfield Emulsions.
- **On-Site Solution Preparation:** Dissolve 1 kg powder per 1.0–1.2 L water to prepare a 45–50% liquid dosing stock. Ships as dry solid powder, saving ~35–40% in freight compared to liquid tankers.
- **Typical STP Dosing Benchmarks (Site-Specific):** Preventative sewer/wet-well dosing: 15–35 mg/L ($g/m^3$) to maintain ORP > -50 mV; Curative sulfide treatment: ~3.0–4.5 kg per kg of dissolved sulfide ($S^{2-}$); Sludge thickeners: 200–400 $g/m^3$.
- **Storage & Handling:** Hygroscopic & oxidising solid (UN 1454, Class 5.1, PG III). Store sealed in a cool, dry warehouse away from moisture and combustible materials. Shelf life: 36 months.

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